NBA

Author: Narek Sahakyan

```
players_data <- read.csv("../data/players_stats_by_season_full_details.csv")

players_data_pr <- players_data %>%
  select(Season, Player, MIN, STL, TOV, ORB, DRB, PF, BLK, PTS) %>%
  arrange(Player, Season)  %>%
  mutate(PR = STL - TOV)

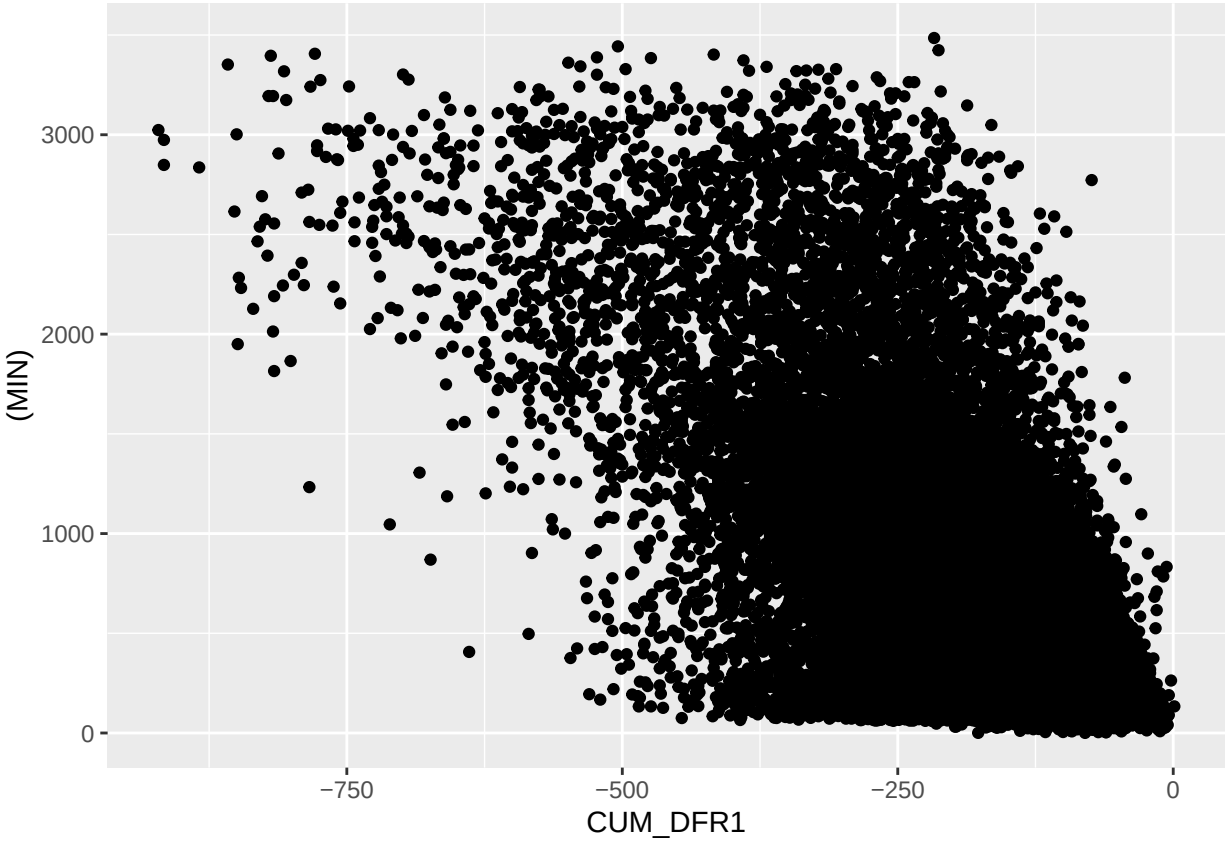
#DFR = defence rating

players_data_diff <- players_data_pr %>%
  arrange(Player, Season)  %>%
  group_by(Player) %>%
  mutate(CUM_STL = rollsum(STL, k = 2, fill = NA, align="right"),
         CUM_TOV = rollsum(TOV, k = 2, fill = NA, align="right"),
         CUM_DRB = rollsum(DRB, k = 2, fill = NA, align="right"),
         CUM_PF = rollsum(PF, k = 2, fill = NA, align="right"),
         CUM_DFR = CUM_STL - CUM_PF,
         CUM_DFR1 = CUM_DFR - CUM_TOV,
         CUM_POSS_RATING = CUM_STL - CUM_TOV
         ) %>%
  drop_na() %>%
  ungroup()


cor(players_data_diff$CUM_DFR1, players_data_diff$MIN)
```

```
## [1] -0.6145041
```

```
ggplot(players_data_diff, aes(x = CUM_DFR1, y = (MIN))) +
  geom_point()
```




```
summary(lm(MIN ~ (CUM_DFR1 + CUM_DFR + CUM_POSS_RATING), data = players_data_diff))
```

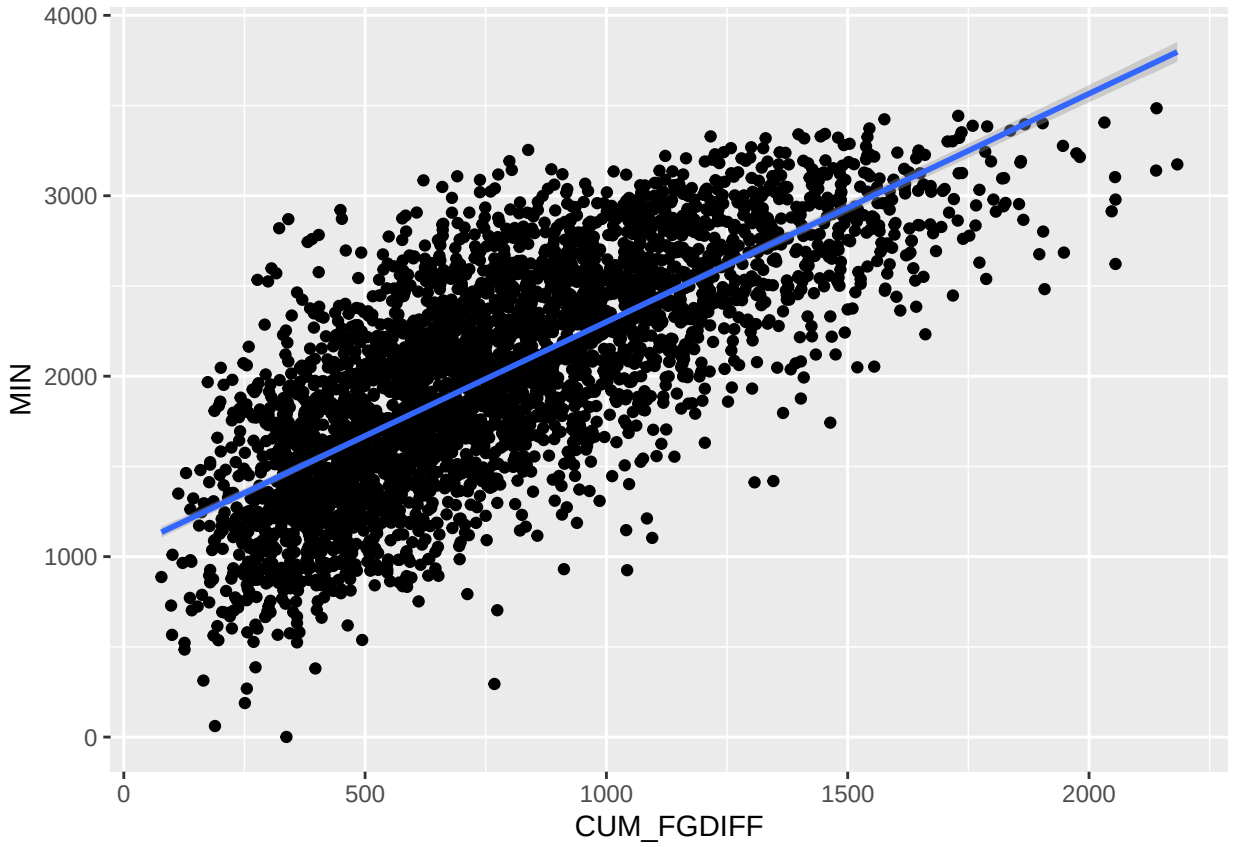
```
##
## Call:
## lm(formula = MIN ~ (CUM_DFR1 + CUM_DFR + CUM_POSS_RATING), data = players_data_diff)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1911.81  -205.79   -22.75   166.26  2223.32
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     20.49029    4.78756    4.28 1.87e-05 ***
## CUM_DFR1        -7.70564    0.05906 -130.48  < 2e-16 ***
## CUM_DFR          5.26631    0.07255   72.59  < 2e-16 ***
## CUM_POSS_RATING  5.28869    0.08945   59.12  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 409.4 on 39363 degrees of freedom
## Multiple R-squared:  0.4601, Adjusted R-squared:  0.4601
## F-statistic: 1.118e+04 on 3 and 39363 DF,  p-value: < 2.2e-16
```

```
library(tidyr)
players_data <- players_data %>%
  filter(League == 'NBA', Stage == 'Regular_Season')
```

```
players_data_pr_2 <- players_data %>%
  select(Season, Player, GP, PTS,  MIN, FGM, FGA, X3PM, X3PA, FTM, FTA)

players_data_diff_2 <- players_data_pr_2 %>%
  arrange(Player, Season)  %>%
  group_by(Player) %>%
  mutate(CUM_FGM = rollsum(FGM, k = 2, fill = NA, align="right"),
         CUM_FGA = rollsum(FGA, k = 2, fill = NA, align="right"),
         CUM_X3PM = rollsum(X3PM, k = 2, fill = NA, align="right"),
         CUM_X3PA = rollsum(X3PA, k = 2, fill = NA, align="right"),
         CUM_FTM = rollsum(FTM, k = 2, fill = NA, align="right"),
         CUM_FTA = rollsum(FTA, k = 2, fill = NA, align="right"),
         CUM_FGDIFF = CUM_FGA - CUM_FGM,
         CUM_X3PMDIFF = CUM_X3PA - CUM_X3PM,
         CUM_FTDIFF = CUM_FTA - CUM_FTM) %>%
  drop_na() %>%
  ungroup()


ggplot(players_data_diff_2, aes(x = CUM_FGDIFF, y = MIN)) +
  geom_point() +
  geom_smooth(method = "lm")
```




```
cor(players_data_diff_2$CUM_FGDIFF, players_data_diff_2$MIN)
```

```
## [1] 0.7380171
```

```r
summary(lm(MIN ~ CUM_FGDIFF + CUM_X3PMDIFF + CUM_FTDIFF,
           data = players_data_diff_2))
```

```
##
## Call:
## lm(formula = MIN ~ CUM_FGDIFF + CUM_X3PMDIFF + CUM_FTDIFF, data = players_data_diff_2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1623.70  -276.44    -4.02   283.11  1204.48
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  993.19290   16.43873   60.42  < 2e-16 ***
## CUM_FGDIFF     1.22794    0.02869   42.81  < 2e-16 ***
## CUM_X3PMDIFF  -0.23905    0.05266   -4.54 5.82e-06 ***
## CUM_FTDIFF     1.24498    0.10871   11.45  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 402.9 on 3580 degrees of freedom
## Multiple R-squared:  0.578,  Adjusted R-squared:  0.5777
## F-statistic:  1635 on 3 and 3580 DF,  p-value: < 2.2e-16
```